STSELab Research Task for *Mostafizur Rahman Khan* – Understanding LLM Explanations for Requirement Quality Violations

Due: Monday, 2nd of November, 2026, 8:00 AM EST

Background

In an externally funded research project, we are studying how well large language models (LLMs) can evaluate the quality of engineering requirements. We provided an LLM with a representative Requirement Set and asked it to evaluate each requirement against the seven quality criteria listed in Table 1. These criteria can be traced back to INCOSE's Systems Engineering Handbook. The ID shown for each criterion is the identifier the LLM uses in its output.

Table 1. Quality Evaluation Criteria for Individual Requirements

Criterion (ID)	Definition
Necessary (A2)	The requirement statement defines capability, characteristic, constraint, or quality factor needed or required to satisfy a lifecycle concept, need, source, or higher-level requirement.
Appropriate (A3)	The specific intent and amount of detail of the need or requirement statement is appropriate to the level (the level of abstraction, organization, or system architecture) of the entity to which it refers.
Unambiguous (A4)	Need and requirement statements must be stated such that their intent is clear and can be interpreted in only one way by all intended audiences.
Complete (A5)	The need statement sufficiently describes the necessary capability, characteristic, constraint, conditions, or quality factor to meet the lifecycle concept or source from which it was transformed. The requirement statement sufficiently describes the necessary capability, characteristic, constraint, conditions, or quality factor to meet the need, source, or higher-level requirement from which it was transformed.
Singular (A6)	The need or requirement statement should state a single capability, characteristic, constraint, or quality factor.
Correct (A9)	The need statement must be an accurate representation of the lifecycle concept or source from which it was transformed. The requirement statement must be an accurate representation of the need, source, or higher-level requirement from which it was transformed.
Conforming (A10)	Statements and expressions of individual needs and requirements should conform to an approved standard pattern and style guide or standard for writing and managing needs and requirements.

We ran this evaluation 100 times, independently, on the same requirement set, using the same LLM with identical settings. Each run produced its own set of flags, and each flag comes with the LLM's written explanation of why it considers that criterion violated.

Data

The output of all 100 runs is provided in the Excel workbook **raw_data.xlsx**, which accompanies this document. It is organized as follows:

- **Worksheets.** The workbook contains 100 worksheets, one per run, named Execution999 through Execution1099.
- **Cell contents.** Each worksheet has a single column with the header "Raw Data" in cell A1. Cell A2 contains that run's complete output as one JSON object.
- **JSON structure.** The JSON object has one key per requirement ID. There are 13 requirements: FR.1–FR.2, PR.1–PR.7, ER.1–ER.3, and RR.1. Each key maps to a list of the criteria the LLM flagged as violated for that requirement. Each entry in the list has three fields: "criterion" (the criterion ID, e.g., A4), "name" (the criterion name, e.g., Unambiguous), and "explanation" (the LLM's written explanation of why it considers that criterion violated).
- **Unflagged criteria.** If a criterion does not appear in a requirement's list, the LLM did not flag that criterion for that requirement in that run.

An abbreviated example of the contents of one cell:

```
{"PR.1": [{"criterion": "A4", "name": "Unambiguous", "explanation": "It is unclear
whether the ±2.5 mm accuracy applies only to the final destination position or to
the entire travel path..."}, {...}], "PR.2": [...], ...}
```

Objective

The objective of this task is to *understand the reasons behind the LLM's requirement quality violation flags*. You will analyze the LLM's explanations to identify what reasons it gives, how consistent those reasons are, and how well they match the intent of each criterion. Please note that LLMs make mistakes, so a flagged violation should not automatically be treated as a confirmed requirement defect.

In this document, an *explanation* is the text the LLM wrote for a single flag, and a *reason* (or rule) is a category you derive from a group of explanations that share the same underlying logic.

We are specifically interested in a better understanding of the following questions:

1. What distinct reasons (or underlying rules) does the LLM give for flagging violations under each quality criterion?
2. How consistently do these reasons appear across requirements and across repeated runs (worksheets), and where and how do they vary?
3. Are the LLM's explanations aligned with what each quality criterion is meant to capture, or does the LLM stretch, merge, or misapply criteria?
4. Do similar reasons appear under different criteria, and what does this suggest about how the LLM distinguishes those criteria?

These questions are a starting point, not a checklist. If the data points to something more important, follow it and tell us.

Deliverables

1. A detailed report and a presentation, both covering:
    - **Research design:** how you got from the raw data to your results, and why you chose this design.
    - **Set of reasons:** the reasons (or rules) behind the LLM's violation flags. Each reason should be clearly defined, distinct from the others, and backed by evidence from the data.
    - **Findings:** what these reasons tell us about how the LLM evaluates requirement quality, including its strengths, weaknesses, consistency, and *any surprises*.
    - **Limitations and open questions:** what your analysis cannot tell us, and what you would investigate next.
2. All data, code, and working files you produced, so we can trace your conclusions back to raw_data.xlsx.